

Saudi Mais Co,for Medical Products Work Instruction

Title

(Chapter.04)

Production & Test Procedure for Sterile Syringes

Doc. #	:	WL-8-2-2-3
Rev. #	:	7
Date.#	:	01.09.2023
Page.#	:	1 / 5

1. **Purpose:**

To specific the production and test procedure for sterile syringes with and without needle to ensure producing good quality product before releasing to the market.

2. **Scope:**

This procedure is applicable for all types of syringes with or without needle produced by Saudi mais company.

3. **Definitions:**

Syringe : A medical device is used for blood collection, extraction, administration of drugs and medicine to the patient.

4. **Responsibilities:**

- Quality Controller are responsible for sampling & testing.
- Quality control supervisors are responsible for checking the obtained results.
- Machine operators and assembly peoples are responsible to produced good quantity and good quality products.

5. **Procedure:**

5.1 **Receiving Bulk Product:**

- Ensure the receiving slip from supplier is correct product as mentioned in purchase order.
- Inspect the carton condition, must be in good condition, no damaged at all cartons with products.
- Take samples & inspect in order to AQL normal inspection procedure for visual inspection, and AQL special inspection method for performance testing and record the results in specified form.
- **Syringe Barrel**
Syringe barrel should be clear and transparent, with printed graduation scale as per syringe capacity, and should not crack, damaged and leakage.
- **Barrel Graduation Scale**
Graduation scale should be Clear and bold printed, visible easy to identify volume.
- **Syringe Plunger**
Syringe plunger Should be clear and smooth surface, easy movement of plunger inside barrel, and should not damaged and crack.
- **Plunger Rubber**
Rubber should be Soft and black colour good fitting with plunger, and should be free from excess lubrication, loose and cracks.

Record the results into OBM inspection report.

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Page. #	:	2 / 5

5.2 Inspection and test report for Syringe

Product Description	Requirement	Status
Packing Condition	Good & without damage	Visual
Appearance	Without any damage	Visual / physical
Type	LL , LT , Catheter tip , Side nozzle	Visual
Free from Irregularities	No Flashing, No Short shot	Visual
Graduation Scale	Printed clearly & accurate as per required sizes	Visual
Rubber Piston Status	Good fitting / without damage	Visual
Nozzle Tip status	No flashing , Accurate hole , No blockage Good luer fitting	Visual
Plunger & Barrel Wings Status	Without any damage / No Crack	Visual
Testing for Liquid leakage	Should not be leak	Visual / Physical
Testing for Air During Aspiration	Should not be leak	Visual / Physical
Testing for Liquid Spillage	Should not be leak	Visual / Physical
Dead Space Determination	As per specification	Visual / Physical
Lubrication Status	Lubricated gasket, no bubbles, no colour change	Visual / Physical

5.3 Visual Inspection:

When inspected by normal or corrected to normal vision the surface of the syringe which come in contact with injection fluids during normal use shall be free from particles and extraneous matter.

5.4 Dimension Control of Syringes

Normal Size of syringe (ml)	Minimum length of scale (mm)	Scale Interval (ml)	Maximum Dead Space (ml)	Tolerance on any graduated capacity equal to or greater then half nominal capacity (%)	Force for leakage Testing Axial pressure gauge Kpa (±5%)
1 ml	57 mm	0.05	0.07	± 5 %	300
2 ml	27 mm	0.2	0.07	± 5 %	300

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Page.#	:	4 / 5

- Isolate the syringe and manometer assembly by means of vacuum tight valve.
- Observe the manometer reading for 60 ± 5 second and record any fall in the reading.
- Examine the syringe to determine if the piston become detached from plunger.

5.8 Testing for Liquid Leakage Under Compression:

- Draw into syringe volume of water exceeding the nominal capacity fo the syringe.
- Expel air and adjust the volume of water in the syringe to the nominal capacity.
- Seal the syringe nozzle with steel female conical fitting connector.
- Apply axial force as syring capacity, force given in dimesion table and maintain the pressure for 30 ± 5 sec.
- Examine the syringe for leakage of water past the piston seals.

5.11 PH determination:

- Using aseptic technique, fill at least three syringes to the nominal capacity with freshly prepared sterile distilled water & maintain at $37 \pm 3^\circ\text{C}$ for 8 hrs.
- Eject the contents & combine them in a vessel made of Borosilicate glass, then determine the pH using pH meter.
- Prepare the control by reserving an aliquot of unused sterile distilled water.
- The pH of the syringe shall be within one unit of that of control fluid.

5.12 Pyrogen Test:

Pyrogen test perfomed as per WL-7-5-1-9.

6. Packaging and Labeling:

- Production controller is responsible to issue production order for packaging.
- Machine operator is responsible to fix die with approved label and set approved parameters for packaging.
- Each product package in blister pack, the label contains the following.
- Product name and code no., reference no., size, lot no., type, temperature, logo, sterilization logos, production date, expiry date, etc.
- Quality controller will responsible to get the packed sample as per AQL procedure and make sure that package forming, sealing, printing and information are correct.
- Every 2 hours quality controller will responsible to get sample as per AQL sampling plan and record the results in in process control form.
- Then to pack in specified small boxes, and arrange in big carton.

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Page.#	:	5 / 5

- The big carton must be with label and detail information.
- 7. Sterilization:**
 - The finished products are sterilized as per work instruction WI-7-5-1-1
- 8. Final Inspection and Product Release**
 - The Quality controller must be check, the unloaded products after sterilization and make final inspection and release.
 - Quality controller will responsible to take sample as per AQL general inspection sampling plan for visual inspection and AQL special inspection for performance testing and inspect the products as requiremnts mention in final inspection and release.
 - The quality assurance manager or quality control manager have to review the completed cycle parameters and inspection reports both documents have to be sign.
 - After confirmation all the test from the Lab, to release the sterilized products, the document once again have to review the quality control manager or quality assurance manager then sign.
 - The production department responsible to record the quantity of products before transferring to the finished goods ware-house and then it could be released to the customer.
- 9. Sampling**
 - Inspection & Test report for Syringe : F-823-T61
 - In-process control for Packaging machine : F-823-i4
 - Final inspection and Release for syringe : F-823-F3
- 10. Reference**

BS EN ISO 7886 - 1

11. Modification / Documentation change History

Date	Changes	Rev.
01.09.2018	Points 5.2 , 8 added	5
15.12.2022	Procedure updated	6
01.09.2023	5.1, 5.3,6 and 8 updated	7

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